

PEER REVIEW

Numerical solution of the SIR epidemic model using Euler and Runge-Kutta Methods GROUP 12

The abstract clearly introduces the topic of ODE, SIR, and identifies the research objective, namely, the use of numerical methods to solve Ordinary Differential Equations (ODEs). The comparison between Euler, RK2, and RK4 methods is well presented, and we can find the main results at the end, particularly regarding the trade-off between numerical accuracy and computational cost. The connection between ODEs and the SIR epidemic model is mentioned; however, since the SIR model appears to be one of the central focuses of the paper, it would be better to provide a more detailed explanation of its role in the abstract. A brief clarification of why the SIR model is particularly relevant.

The introduction is also clear and well-structured. It prepares us for the rest of the paper by explaining the importance of ODEs and showing the relevance of numerical methods when analytical solutions are unavailable. The research problem is clearly stated, and the motivation is convincing, especially in the context of epidemiological modelling and public health applications. The discussion of numerical methods is interesting and useful. And we understand how it will help with the SIR model. However, a stronger and more explicit connection between ODEs and the specific application to the SIR model would improve coherence with the title and the main subject. Since the SIR model is one of the main subjects of the paper, introducing it earlier and with slightly more detail would help guide the reader more effectively.

We felt that the paper's explanation of Euler's method (3.3) was well articulated, however we would have benefited from more detail on the theory behind the Runge-Kutta methods (3.4). We could see that by averaging the slope over an interval, the approximation is more accurate (3.4.1), though we were not able to identify why this is the case. That said, the paper does cohesively explain the error for the Runge-Kutta second-order method (RK2) (3.4.2), with respect to Euler's method. Similarly, for the Runge-Kutta fourth-order method (RK4) (3.4.4), the error analysis is clear, though we felt the explanation of the method itself could be elaborated upon to enhance understanding.

The numerical methods were clearly explained, and we felt that the flow charts and formulas were very well executed, allowing us to understand how the theory extends to numerical implementation. However, although briefly touched upon in the Appendix (9.6), we couldn't quite make out how these methods could be applied to a system of ODEs and would be interested to see more exploration into how accuracy and error are affected by approximating an entire system.

Additionally, the article compares the Euler, RK2 and RK4 methods clearly, starting from a simple benchmark example and then showing the SIR model (5.1). The article shows that for computing accuracy, RK4 is better than RK2 and even more so than Euler's method. However, there is a small limitation in the SIR section (5.3). Only RK4 is used for the simulation. Since the main goal is to compare the three methods, it would be more convincing and clearer if Euler and RK2 were also included in the SIR example. Additionally, the graphs are clear and abundant, but there can be more explanations to the graphs so the idea would help present the results more comprehensively.

We found the analysis of the accuracy and error for each of the implemented numerical methods to be easy to follow and well-reasoned. We greatly appreciated the use of graphs to demonstrate the accuracy and convergence of error (5.1). They allowed for visual confirmation of the theory presented earlier in the paper, that the accuracy is improved for higher order methods.

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Although the paper neatly demonstrates a numerical approximation for the SIR model (5.3), we weren't able to distinguish exactly how the numerical methods have been used to solve this system. We would have also been interested to read more about how numerical analysis can be used to tell us more about the SIR model, such as how the initial conditions and the transmission and recovery rates can affect the stability of the SIR model.

Overall, we found the paper to be very interesting and well written. We believe it has demonstrated how useful numerical analysis can be for real-world applications, especially the SIR model. We also believe that with some minor adjustments this paper would, if hypothetically being graded for a class, achieve top marks.